



IoT for health insurance: IoT-based devices help establish transparency between insurance companies and customers. They help to document the cost of treating the health issue, as well as submit and process claims. They enable insurance companies to validate claims through the data captured by these devices.

According to a report by *Markets and Market*, the IoT technology industry is projected to grow from US\$ 945.6 billion in 2024 to US\$ 1,377.8 billion by 2029, registering a CAGR of 7.8% during this period. As per a *Future Market Insights* report, the global healthcare IoT market is projected to grow from US\$ 108.6 billion in 2024 to US\$ 167.7 billion by 2028, at a CAGR of 11.47%.

Challenges in implementing IoT in healthcare

Scalability: As the number of connected devices increases, so does the volume of data. This requires robust cloud infrastructure, edge computing, and device lifecycle management. Ensuring that the IoT devices can scale to handle this data efficiently is a significant challenge.

Interoperability: IoT devices often use a variety of proprietary protocols and standards. Ensuring seamless interoperability between these diverse systems is a challenge. Lack of universal standards makes integration with EHRs (electronic health records), hospital systems and hospital infrastructure difficult. For example, a smart pill dispenser may not easily sync with a hospital's Epic or Cerner EHR system.

Security: Sensitive patient data is continuously transmitted across devices and networks. IoT systems are vulnerable to cyber threats that include data breaches, unauthorised access and ransomware attacks. Ensuring robust security measures within the IoT middleware to protect data and devices is crucial but challenging.

Privacy concerns: Safeguarding patients' data and ensuring compliance with privacy regulations like HIPAA and GDPR is an ongoing challenge. For example, a wearable glucose monitor transmitting data to the cloud must ensure encryption at rest and in transit.

Reliability: IoT devices require stable, low-latency connections (Wi-Fi, 5G, LPWAN). In rural or poor network settings, connectivity gaps can disrupt monitoring. For example, remote patient monitoring for cardiac patients may fail if network coverage is weak.

Characteristics of IoT healthcare systems

To perform well, IoT healthcare systems must possess a few key characteristics.

Connectivity and interoperability: There should be seamless communication between devices, gateways, the cloud, and hospital systems (EHR/EMR). HL7, FHIR, and DICOM are the standards that ensure interoperability across vendors and platforms.

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